

Midterm Review 1

CS 185/285

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Part 1:

Imitation learning

1. Collect data by asking a person to make demonstrations

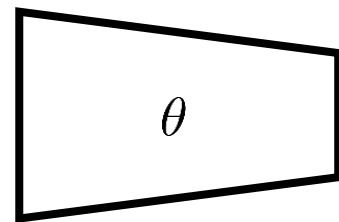
$$\{(\mathbf{o}_1^{(i)}, \mathbf{a}_1^{(i)}, \dots, \mathbf{o}_H^{(i)}, \mathbf{a}_H^{(i)})\}_{i=1}^N$$

2. Run supervised learning (a.k.a. maximum likelihood estimation) on this data with your favorite neural network

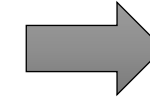
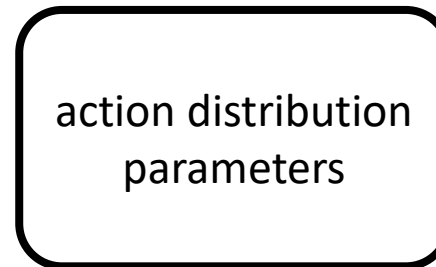
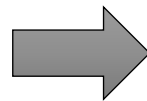
$$\arg \max_{\theta} \sum_{i=1}^N \sum_{t=1}^H \log \pi_{\theta}(\mathbf{a}_t^{(i)} | \mathbf{o}_t^{(i)})$$



$\mathbf{o}_t$



encoder
(e.g., ViT, ResNet)



sample



$\mathbf{a}_t$

Quantifying how bad it could be

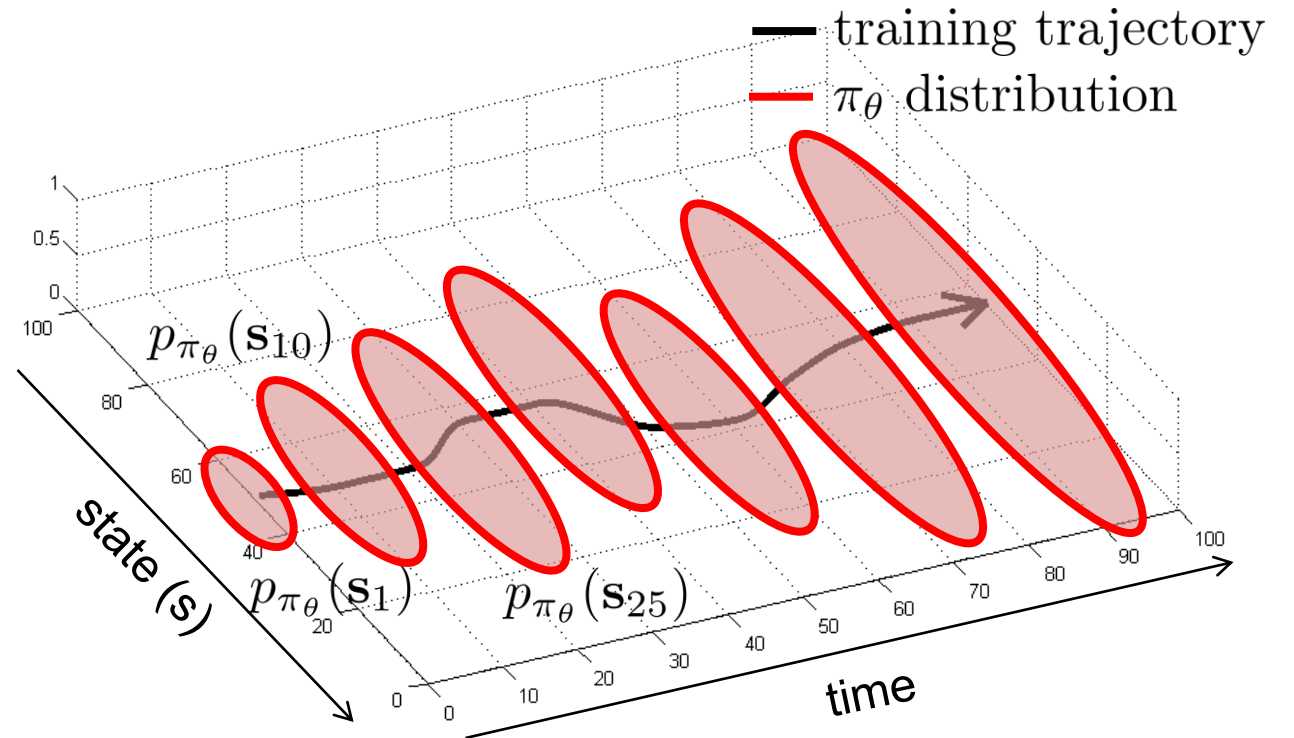
$$\sum_{t=1}^H \mathbb{E}_{\mathbf{a}_t \sim \pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t), \mathbf{s}_t \sim p_{\pi_{\theta}}(\mathbf{s}_t)} [c(\mathbf{s}_t, \mathbf{a}_t)]$$

note that this depends
on past actions

This sum corresponds to the
expected total number of mistakes

Question: how does the total number of mistakes scale with the length of the trajectory (the “horizon”)

$$c(\mathbf{s}_t, \mathbf{a}_t) = \begin{cases} 0 & \text{if } \mathbf{a}_t = \pi^*(\mathbf{s}_t) \\ 1 & \text{otherwise} \end{cases}$$



A more general analysis

assume: $\mathbb{E}_{\mathbf{s}_t \sim p_{\text{train}}(\mathbf{s}_t)} [\pi_{\theta}(\mathbf{a}_t \neq \pi^*(\mathbf{s}_t) | \mathbf{s}_t)] \leq \epsilon$

“if we sample a state from the training distribution, we are unlikely to make a mistake”

A useful property:

$$p_{\pi_{\theta}}(\mathbf{s}_t) = \underbrace{(1 - \epsilon)^t}_{\text{probability we made no mistakes}} p_{\text{train}}(\mathbf{s}_t) + \underbrace{(1 - (1 - \epsilon)^t)}_{\text{some } \textit{other} \text{ distribution}} p_{\text{mistake}}(\mathbf{s}_t)$$

A more general analysis

Total variation divergence $D_{\text{TV}}(p, q) = \frac{1}{2} \sum_x |p(x) - q(x)| \leq 1$

$$p_{\pi_\theta}(\mathbf{s}_t) = (1 - \epsilon)^t p_{\text{train}}(\mathbf{s}_t) + (1 - (1 - \epsilon)^t) p_{\text{mistake}}(\mathbf{s}_t)$$

$$D_{\text{TV}}(p_{\text{train}}, p_{\pi_\theta}) = \underbrace{\frac{1}{2} \sum_{\mathbf{s}_t} |p_{\text{train}}(\mathbf{s}_t) - p_{\pi_\theta}(\mathbf{s}_t)|}_{\substack{\frac{1}{2} \sum_{\mathbf{s}_t} |p_{\text{train}}(\mathbf{s}_t) - (1 - \epsilon)^t p_{\text{train}}(\mathbf{s}_t) - (1 - (1 - \epsilon)^t) p_{\text{mistake}}(\mathbf{s}_t)| \\ \uparrow \\ (1 - \epsilon)^t p_{\text{train}}(\mathbf{s}_t) + (1 - (1 - \epsilon)^t) p_{\text{train}}(\mathbf{s}_t) \\ = (1 - (1 - \epsilon)^t) \frac{1}{2} \sum_{\mathbf{s}_t} |p_{\text{train}}(\mathbf{s}_t) - p_{\text{mistake}}(\mathbf{s}_t)|}}$$

useful identity:

$$(1 - \epsilon)^t \geq 1 - \epsilon t \text{ for } \epsilon \in [0, 1] \qquad \leq (1 - (1 - \epsilon)^t) \leq \epsilon t$$

A more general analysis

$$\begin{aligned} & \sum_{t=1}^H \mathbb{E}_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t | \mathbf{s}_t), \mathbf{s}_t \sim p_{\pi_\theta}(\mathbf{s}_t)} [c(\mathbf{s}_t, \mathbf{a}_t)] \\ &= \sum_{t=1}^H \sum_{\mathbf{s}_t} p_{\pi_\theta}(\mathbf{s}_t) \mathbb{E}_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t | \mathbf{s}_t)} [c(\mathbf{s}_t, \mathbf{a}_t)] \\ &= \sum_{t=1}^H \sum_{\mathbf{s}_t} (p_{\text{train}}(\mathbf{s}_t) + p_{\pi_\theta}(\mathbf{s}_t) - p_{\text{train}}(\mathbf{s}_t)) \mathbb{E}_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t | \mathbf{s}_t)} [c(\mathbf{s}_t, \mathbf{a}_t)] \\ &\leq \sum_{t=1}^H \sum_{\mathbf{s}_t} (p_{\text{train}}(\mathbf{s}_t) + |p_{\pi_\theta}(\mathbf{s}_t) - p_{\text{train}}(\mathbf{s}_t)|) \mathbb{E}_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t | \mathbf{s}_t)} [c(\mathbf{s}_t, \mathbf{a}_t)] \\ &= \sum_{t=1}^H \left[\sum_{\mathbf{s}_t} p_{\text{train}}(\mathbf{s}_t) \mathbb{E}_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t | \mathbf{s}_t)} [c(\mathbf{s}_t, \mathbf{a}_t)] \right] + \left[\sum_{\mathbf{s}_t} |p_{\pi_\theta}(\mathbf{s}_t) - p_{\text{train}}(\mathbf{s}_t)| \mathbb{E}_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t | \mathbf{s}_t)} [c(\mathbf{s}_t, \mathbf{a}_t)] \right] \end{aligned}$$

A more general analysis

$$\begin{aligned}
 & \sum_{t=1}^H \mathbb{E}_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t | \mathbf{s}_t), \mathbf{s}_t \sim p_{\pi_\theta}(\mathbf{s}_t)} [c(\mathbf{s}_t, \mathbf{a}_t)] \\
 &= \sum_{t=1}^H \left[\underbrace{\sum_{\mathbf{s}_t} p_{\text{train}}(\mathbf{s}_t) \mathbb{E}_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t | \mathbf{s}_t)} [c(\mathbf{s}_t, \mathbf{a}_t)]}_{\mathbb{E}_{\mathbf{s}_t \sim p_{\text{train}}(\mathbf{s}_t)} [\pi_\theta(\mathbf{a}_t \neq \pi^*(\mathbf{s}_t) | \mathbf{s}_t)] \leq \epsilon} + \underbrace{\left[\sum_{\mathbf{s}_t} |p_{\pi_\theta}(\mathbf{s}_t) - p_{\text{train}}(\mathbf{s}_t)| \mathbb{E}_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t | \mathbf{s}_t)} [c(\mathbf{s}_t, \mathbf{a}_t)] \right]}_{= 2D_{\text{TV}}(p_{\pi_\theta}, p_{\text{train}}) \leq 2\epsilon t} \underbrace{\leq 1} \right] \\
 &\leq \sum_{t=1}^H \epsilon + 2\epsilon t \quad O(\epsilon H^2)
 \end{aligned}$$

This is bad because error increases *quadratically* with horizon

Fixing distributional shift

can we make $p_{\text{data}}(\mathbf{o}_t) = p_{\pi_\theta}(\mathbf{o}_t)$?

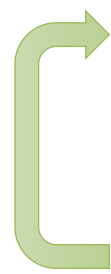
idea: instead of being clever about $p_{\pi_\theta}(\mathbf{o}_t)$, be clever about $p_{\text{data}}(\mathbf{o}_t)$!

DAgger: Dataset Aggregation

goal: collect training data from $p_{\pi_\theta}(\mathbf{o}_t)$ instead of $p_{\text{data}}(\mathbf{o}_t)$

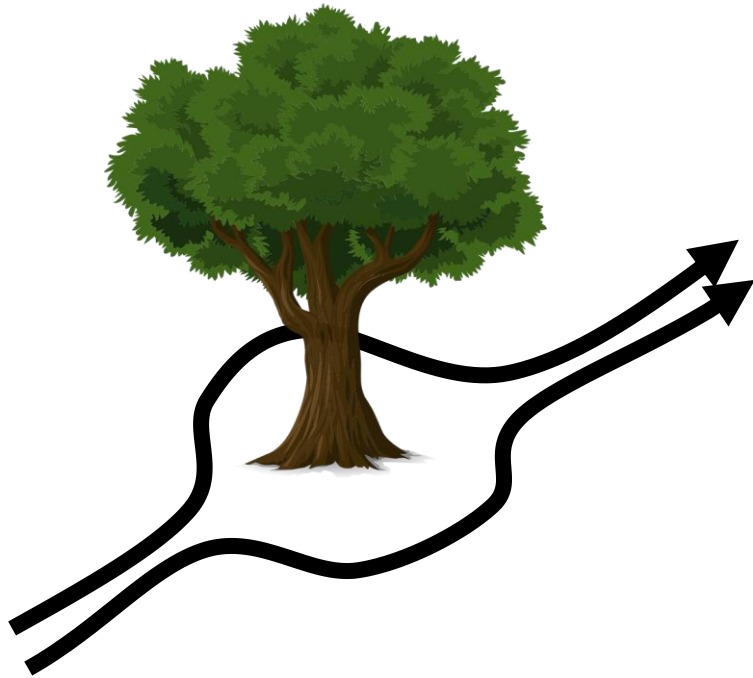
how? just run $\pi_\theta(\mathbf{a}_t|\mathbf{o}_t)$

but need labels $\mathbf{a}_t$!

- 
1. train $\pi_\theta(\mathbf{a}_t|\mathbf{o}_t)$ from human data $\mathcal{D} = \{\mathbf{o}_1, \mathbf{a}_1, \dots, \mathbf{o}_N, \mathbf{a}_N\}$
 2. run $\pi_\theta(\mathbf{a}_t|\mathbf{o}_t)$ to get dataset $\mathcal{D}_\pi = \{\mathbf{o}_1, \dots, \mathbf{o}_M\}$
 3. Ask human to label $\mathcal{D}_\pi$ with actions $\mathbf{a}_t$
 4. Aggregate: $\mathcal{D} \leftarrow \mathcal{D} \cup \mathcal{D}_\pi$

Why might we fail to fit the expert?

1. Non-Markovian behavior
2. Multimodal behavior



Autoregressive discretization

Why does this work?

first step: $p(a_{t,0}|\mathbf{s}_t)$

second step: $p(a_{t,1}|\mathbf{s}_t, a_{t,0})$

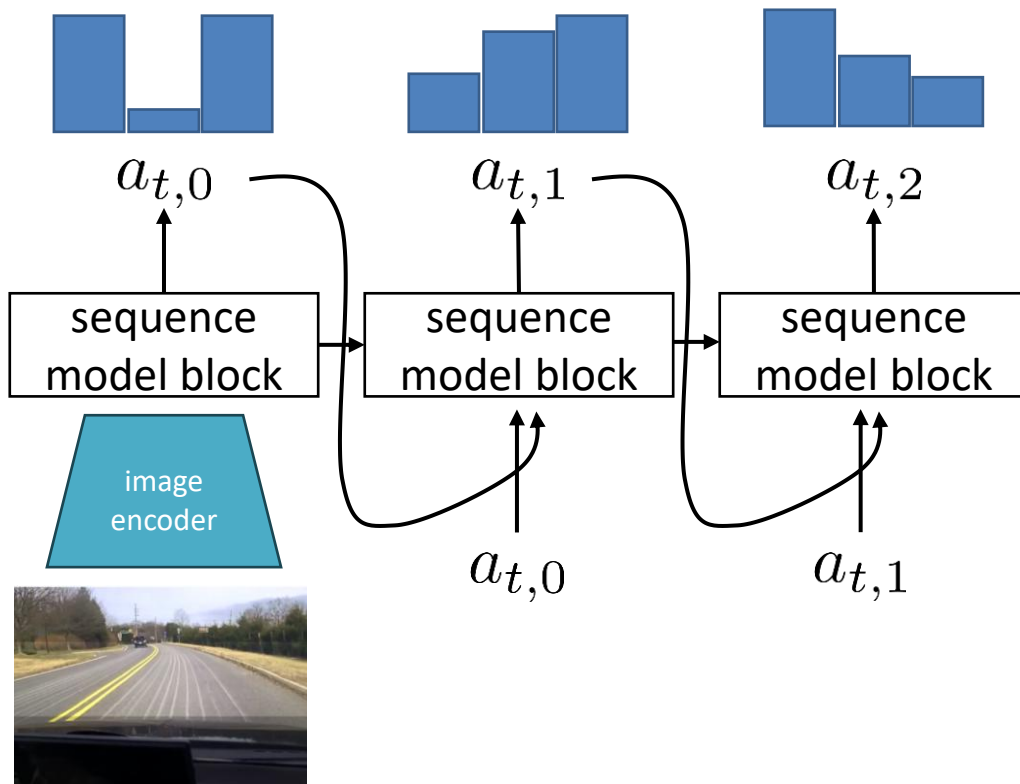
third step: $p(a_{t,2}|\mathbf{s}_t, a_{t,0}, a_{t,1})$

$$p(a_{t,2}|\mathbf{s}_t, a_{t,0}, a_{t,1})p(a_{t,1}|\mathbf{s}_t, a_{t,0})p(a_{t,0}|\mathbf{s}_t)$$

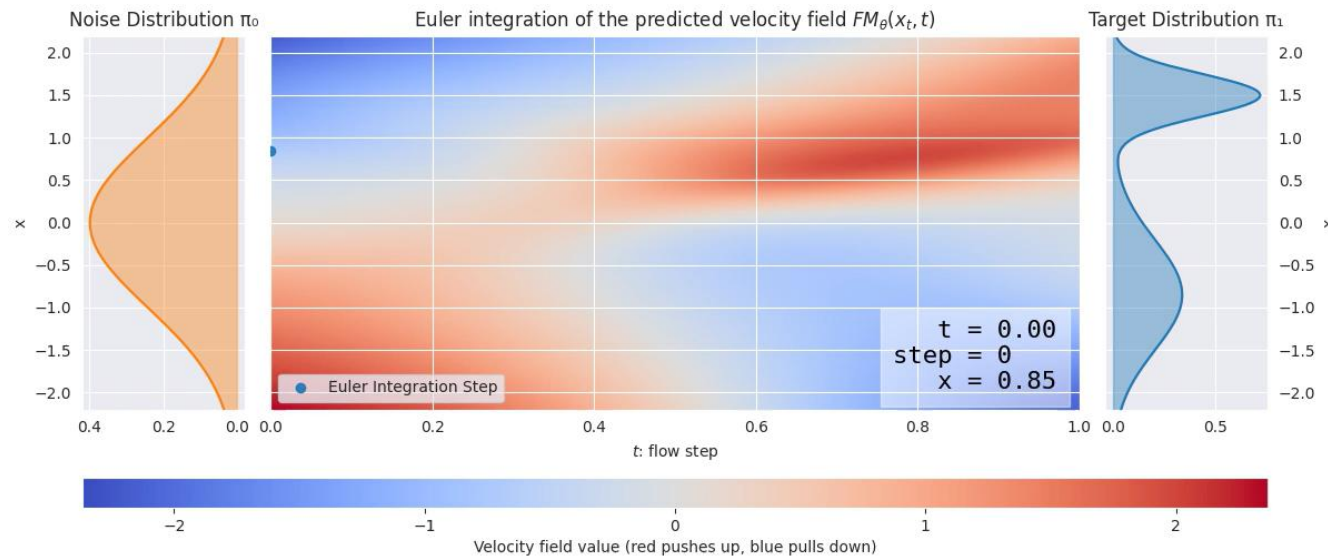
$$= p(a_{t,0}, a_{t,1}, a_{t,2}|\mathbf{s}_t)$$

$$= p(\mathbf{a}_t|\mathbf{s}_t)$$

e.g.,
autoregressive
Transformer



Flow matching overview



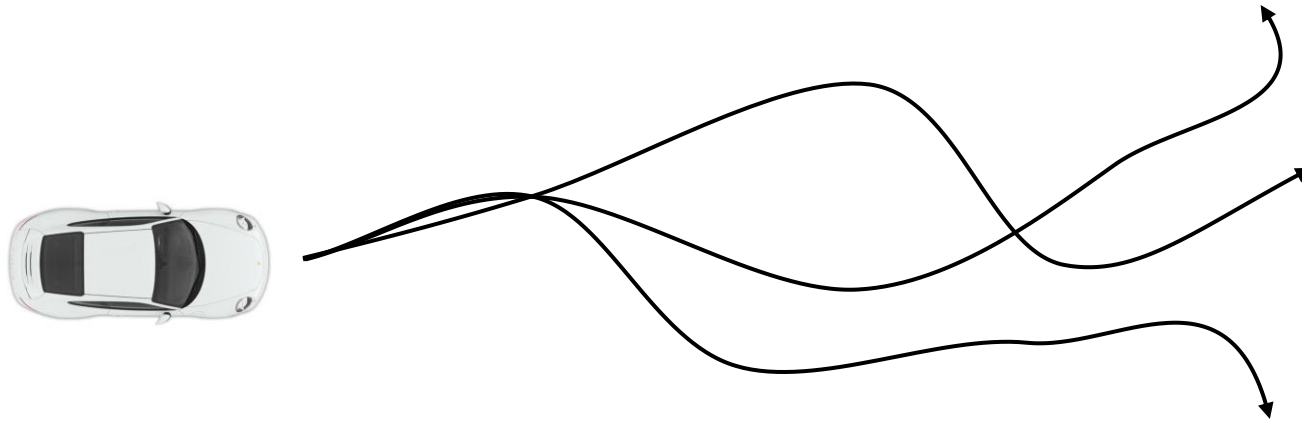
to sample:

1. sample $\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})$
2. integrate
for $t \in \{0, \Delta t, 2\Delta t, 3\Delta t, \dots, 1 - \Delta t\}$:
$$\mathbf{x}_{t+\Delta t} \leftarrow \mathbf{x}_t + v(\mathbf{x}_t, t)\Delta t$$
3. return $\mathbf{x}_1$

to train:

1. sample $\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})$
2. sample $\mathbf{x}_1 \in \mathcal{D}$ where $\mathcal{D} = \{\mathbf{x}^{(i)}\}_{i=1}^N$
3. sample $t \sim p(t)$ (e.g., $p(t) = \mathcal{U}(0, 1)$)
4. compute $\mathbf{x}_t = t\mathbf{x}_1 + (1 - t)\mathbf{x}_0$
5. update v with $\nabla \|v(\mathbf{x}_t, t) - \underbrace{(\mathbf{x}_1 - \mathbf{x}_0)}_{\text{target velocity}}\|^2$

Goal-conditioned behavioral cloning



training time:

demo 1: $\{\mathbf{s}_1, \mathbf{a}_1, \dots, \mathbf{s}_{T-1}, \mathbf{a}_{T-1}, \mathbf{s}_T\}$ ← successful demo for reaching $\mathbf{s}_T$

demo 2: $\{\mathbf{s}_1, \mathbf{a}_1, \dots, \mathbf{s}_{T-1}, \mathbf{a}_{T-1}, \mathbf{s}_T\}$

learn $\pi_\theta(\mathbf{a}|\mathbf{s}, \mathbf{g})$ ← goal state

demo 3: $\{\mathbf{s}_1, \mathbf{a}_1, \dots, \mathbf{s}_{T-1}, \mathbf{a}_{T-1}, \mathbf{s}_T\}$

for each demo $\{\mathbf{s}_1^i, \mathbf{a}_1^i, \dots, \mathbf{s}_{T-1}^i, \mathbf{a}_{T-1}^i, \mathbf{s}_T^i\}$

maximize $\log \pi_\theta(\mathbf{a}_t^i | \mathbf{s}_t^i, \mathbf{g} = \mathbf{s}_T^i)$

Part 2:

Policy gradients

Direct policy differentiation

$$\begin{aligned}
 \theta^* &= \arg \max_{\theta} J(\theta) \\
 J(\theta) &= \mathbb{E}_{\tau \sim p_{\theta}(\tau)} [r(\tau)] \\
 \nabla_{\theta} J(\theta) &= \mathbb{E}_{\tau \sim p_{\theta}(\tau)} [\nabla_{\theta} \log p_{\theta}(\tau) r(\tau)]
 \end{aligned}$$

log of both sides

$$\underbrace{p_{\theta}(\mathbf{s}_1, \mathbf{a}_1, \dots, \mathbf{s}_T, \mathbf{a}_T)}_{p_{\theta}(\tau)} = p(\mathbf{s}_1) \prod_{t=1}^H \pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t) p(\mathbf{s}_{t+1} | \mathbf{s}_t, \mathbf{a}_t)$$

$$\log p_{\theta}(\tau) = \log p(\mathbf{s}_1) + \sum_{t=1}^H \log \pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t) + \log p(\mathbf{s}_{t+1} | \mathbf{s}_t, \mathbf{a}_t)$$

$$\nabla_{\theta} \left[\log p(\mathbf{s}_1) + \sum_{t=1}^H \log \pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t) + \log p(\mathbf{s}_{t+1} | \mathbf{s}_t, \mathbf{a}_t) \right]$$

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\tau \sim p_{\theta}(\tau)} \left[\left(\sum_{t=1}^H \nabla_{\theta} \log \pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t) \right) \left(\sum_{t=1}^H r(\mathbf{s}_t, \mathbf{a}_t) \right) \right]$$

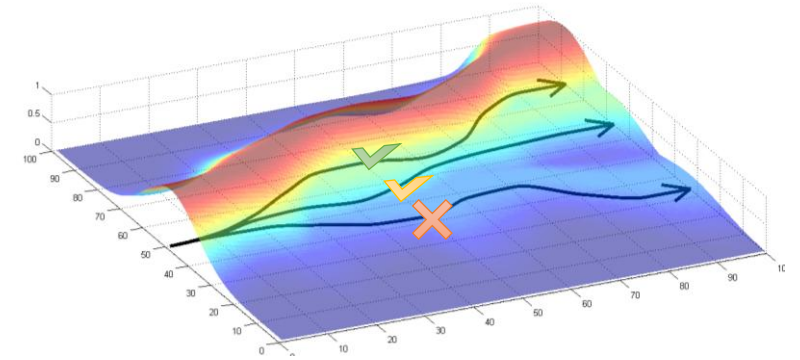
Baselines

a convenient identity

$$p_{\theta}(\tau) \nabla_{\theta} \log p_{\theta}(\tau) = \nabla_{\theta} p_{\theta}(\tau)$$

$$\nabla_{\theta} J(\theta) \approx \frac{1}{N} \sum_{i=1}^N \nabla_{\theta} \log p_{\theta}(\tau) [r(\tau) - b]$$

$$b = \frac{1}{N} \sum_{i=1}^N r(\tau) \quad \text{but... are we *allowed* to do that??}$$



$$\mathbb{E}[\nabla_{\theta} \log p_{\theta}(\tau) b] = \int p_{\theta}(\tau) \nabla_{\theta} \log p_{\theta}(\tau) b d\tau = \int \nabla_{\theta} p_{\theta}(\tau) b d\tau = b \nabla_{\theta} \int p_{\theta}(\tau) d\tau = b \nabla_{\theta} 1 = 0$$

subtracting a baseline is *unbiased* in expectation!

average reward is *not* the best baseline, but it's pretty good!

Off-policy learning & importance sampling

$$\theta^* = \arg \max_{\theta} J(\theta)$$

$$J(\theta) = E_{\tau \sim p_{\theta}(\tau)}[r(\tau)]$$

what if we don't have samples from $p_{\theta}(\tau)$?

(we have samples from some $\bar{p}(\tau)$ instead)

$$J(\theta) = E_{\tau \sim \bar{p}(\tau)} \left[\frac{p_{\theta}(\tau)}{\bar{p}(\tau)} r(\tau) \right]$$

$$p_{\theta}(\tau) = p(\mathbf{s}_1) \prod_{t=1}^H \pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t) p(\mathbf{s}_{t+1} | \mathbf{s}_t, \mathbf{a}_t)$$

$$\frac{p_{\theta}(\tau)}{\bar{p}(\tau)} = \frac{\cancel{p(\mathbf{s}_1)} \prod_{t=1}^H \pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t) \cancel{p(\mathbf{s}_{t+1} | \mathbf{s}_t, \mathbf{a}_t)}}{\cancel{p(\mathbf{s}_1)} \prod_{t=1}^H \bar{\pi}(\mathbf{a}_t | \mathbf{s}_t) \cancel{p(\mathbf{s}_{t+1} | \mathbf{s}_t, \mathbf{a}_t)}} = \frac{\prod_{t=1}^H \pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t)}{\prod_{t=1}^H \bar{\pi}(\mathbf{a}_t | \mathbf{s}_t)}$$

importance sampling

$$\begin{aligned} E_{x \sim p(x)}[f(x)] &= \int p(x) f(x) dx \\ &= \int \frac{q(x)}{q(x)} p(x) f(x) dx \\ &= \int q(x) \frac{p(x)}{q(x)} f(x) dx \\ &= E_{x \sim q(x)} \left[\frac{p(x)}{q(x)} f(x) \right] \end{aligned}$$

A first-order approximation for IS

$$\nabla_{\theta'} J(\theta') = E_{\tau \sim p_{\theta}(\tau)} \left[\sum_{t=1}^H \nabla_{\theta'} \log \pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t) \underbrace{\left(\prod_{t'=1}^t \frac{\pi_{\theta'}(\mathbf{a}_{t'} | \mathbf{s}_{t'})}{\pi_{\theta}(\mathbf{a}_{t'} | \mathbf{s}_{t'})} \right)}_{\text{exponential in } T} \left(\sum_{t'=t}^H r(\mathbf{s}_{t'}, \mathbf{a}_{t'}) \right) \right]$$

let's write the objective a bit differently...

on-policy policy gradient: $\nabla_{\theta} J(\theta) \approx \frac{1}{N} \sum_{i=1}^N \sum_{t=1}^H \nabla_{\theta} \log \pi_{\theta}(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)}) \hat{A}_t^{(i)}$

$(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) \sim \pi_{\theta}(\mathbf{s}_t, \mathbf{a}_t)$

off-policy policy gradient: $\nabla_{\theta'} J(\theta') \approx \frac{1}{N} \sum_{i=1}^N \sum_{t=1}^H \frac{\pi_{\theta'}(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)})}{\pi_{\theta}(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)})} \nabla_{\theta'} \log \pi_{\theta'}(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)}) \hat{A}_t^{(i)}$

this is “a little bit wrong”
we'll see next time why

$$= \frac{1}{N} \sum_{i=1}^N \sum_{t=1}^H \frac{\cancel{\pi_{\theta'}(\mathbf{s}_t^{(i)})}}{\cancel{\pi_{\theta}(\mathbf{s}_t^{(i)})}} \frac{\pi_{\theta'}(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)})}{\pi_{\theta}(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)})} \nabla_{\theta'} \log \pi_{\theta'}(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)}) \hat{A}_t^{(i)}$$

ignore this part

it's mostly ok to do this

Clipped importance weights

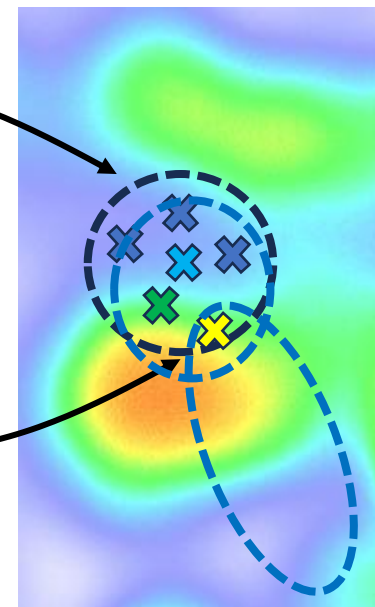
$$J(\theta') = E_{\tau \sim p_{\theta}(\tau)} \left[\overbrace{\frac{p_{\theta'}(\tau)}{p_{\theta}(\tau)}}^{w(\tau)} r(\tau) \right]$$

“surrogate objective”

$$\nabla_{\theta'} J(\theta') = E_{\tau \sim p_{\theta}(\tau)} \left[\frac{p_{\theta'}(\tau)}{p_{\theta}(\tau)} \nabla_{\theta'} \log p_{\theta'}(\tau) r(\tau) \right]$$

$w(\tau)$
keeps
decreasing here

$w(\tau)$
keeps
increasing here



try 0.1

↓

$$\text{redefine } w(\tau) = \max \left\{ 1 - \epsilon, \min \left\{ 1 + \epsilon, \frac{p_{\theta'}(\tau)}{p_{\theta}(\tau)} \right\} \right\}$$

puts a limit on how
much probabilities can
increase or decrease

One more detail

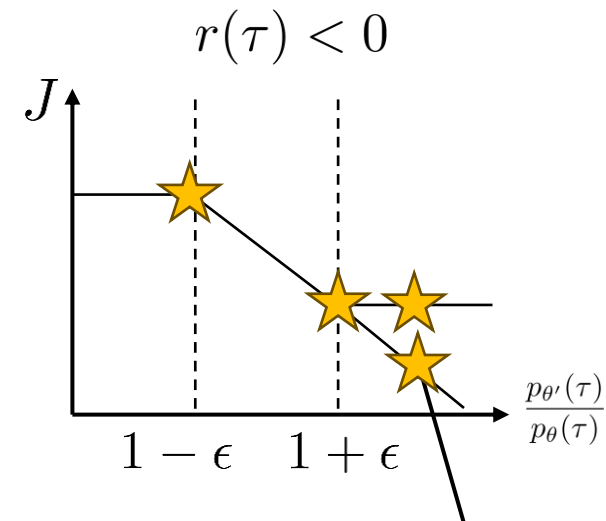
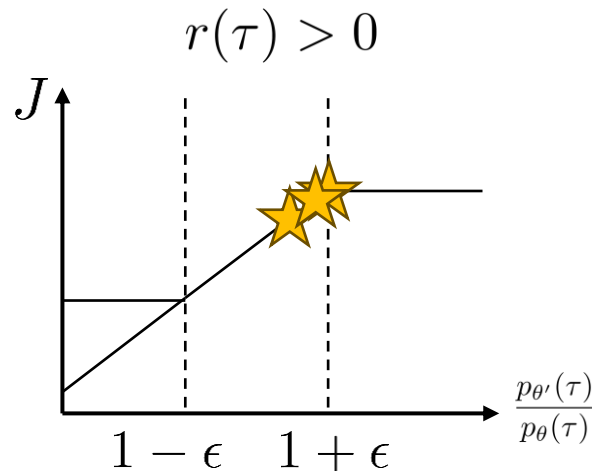
~~$$J(\theta') = E_{\tau \sim p_{\theta}(\tau)} [w(\tau)r(\tau)]$$~~

~~$$w(\tau) = \max \left\{ 1 - \epsilon, \min \left\{ 1 + \epsilon, \frac{p_{\theta'}(\tau)}{p_{\theta}(\tau)} \right\} \right\}$$~~

$$J(\theta') = E_{\tau \sim p_{\theta}(\tau)} [\min\{w(\tau)r(\tau), w_c(\tau)r(\tau)\}]$$

$$w(\tau) = \frac{p_{\theta'}(\tau)}{p_{\theta}(\tau)}$$

$$w_c(\tau) = \max \left\{ 1 - \epsilon, \min \left\{ 1 + \epsilon, \frac{p_{\theta'}(\tau)}{p_{\theta}(\tau)} \right\} \right\}$$



not paying any price
for drifting away from
data here!

Clipped importance sampling policy gradient

policy gradient with clipped IS (PPO):

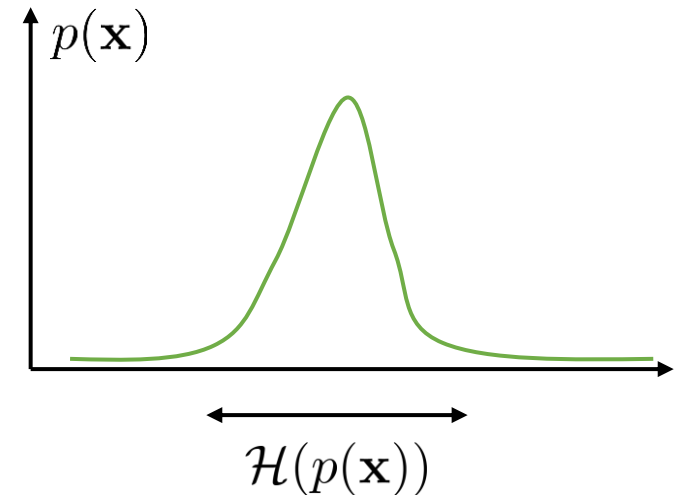
1. sample $\{\tau^{(i)}\}$ from π_θ (run the policy)
2. evaluate $y_t^{(i)} = r(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) + \hat{V}_\phi^\pi(\mathbf{s}_{t+1}^{(i)})$
3. fit $\hat{V}_\phi^\pi(\mathbf{s})$ to targets $\{y_t^{(i)}\}$
4. evaluate $\hat{A}_{\text{GAE}}^\pi(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) = \sum_{t'=t}^{\infty} (\gamma\lambda)^{t'-t} \delta_{t'}^{(i)}$
5. $\theta' \leftarrow \theta$
6. $\nabla_{\theta'} J(\theta') \approx \nabla_{\theta'} \mathcal{L}_{\text{CLIP}}(\theta')$
7. $\theta' \leftarrow \theta' + \alpha \nabla_{\theta'} J(\theta')$
6. $\theta \leftarrow \theta'$

this results in the widely used PPO algorithm

Some more details:

use GAE for $y_t^{(i)}$ as well

add $\sum_i \sum_t \mathcal{H}(\pi_{\theta'}(\cdot | \mathbf{s}_t^{(i)}))$ to objective
“entropy regularization”



$$\mathcal{L}_{\text{CLIP}}(\theta') = \sum_{i=1}^N \sum_{t=1}^H \min \left\{ \frac{\pi_{\theta'}(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)})}{\pi_\theta(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)})} \hat{A}_t^{(i)}, \text{clip} \left(\frac{\pi_{\theta'}(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)})}{\pi_\theta(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)})}, 1 - \epsilon, 1 + \epsilon \right) \hat{A}_t^{(i)} \right\}$$

Policy gradient as **policy iteration**

$$J(\theta) = E_{\tau \sim p_{\theta}(\tau)} \left[\sum_t \gamma^t r(\mathbf{s}_t, \mathbf{a}_t) \right]$$

$$J(\theta') - J(\theta) = J(\theta') - E_{\mathbf{s}_0 \sim p(\mathbf{s}_0)} [V^{\pi_{\theta}}(\mathbf{s}_0)]$$

$$= J(\theta') - E_{\tau \sim p_{\theta'}(\tau)} [V^{\pi_{\theta}}(\mathbf{s}_0)]$$

$$\text{claim: } J(\theta') - J(\theta) = E_{\tau \sim p_{\theta'}(\tau)} \left[\sum_t \gamma^t A^{\pi_{\theta}}(\mathbf{s}_t, \mathbf{a}_t) \right]$$

$$= J(\theta') - E_{\tau \sim p_{\theta'}(\tau)} \left[\sum_{t=0}^{\infty} \gamma^t V^{\pi_{\theta}}(\mathbf{s}_t) - \sum_{t=1}^{\infty} \gamma^t V^{\pi_{\theta}}(\mathbf{s}_t) \right]$$

$$= J(\theta') + E_{\tau \sim p_{\theta'}(\tau)} \left[\sum_{t=0}^{\infty} \gamma^t (\gamma V^{\pi_{\theta}}(\mathbf{s}_{t+1}) - V^{\pi_{\theta}}(\mathbf{s}_t)) \right]$$

$$= E_{\tau \sim p_{\theta'}(\tau)} \left[\sum_{t=0}^{\infty} \gamma^t r(\mathbf{s}_t, \mathbf{a}_t) \right] + E_{\tau \sim p_{\theta'}(\tau)} \left[\sum_{t=0}^{\infty} \gamma^t (\gamma V^{\pi_{\theta}}(\mathbf{s}_{t+1}) - V^{\pi_{\theta}}(\mathbf{s}_t)) \right]$$

$$= E_{\tau \sim p_{\theta'}(\tau)} \left[\sum_{t=0}^{\infty} \gamma^t (r(\mathbf{s}_t, \mathbf{a}_t) + \gamma V^{\pi_{\theta}}(\mathbf{s}_{t+1}) - V^{\pi_{\theta}}(\mathbf{s}_t)) \right]$$

$$= E_{\tau \sim p_{\theta'}(\tau)} \left[\sum_{t=0}^{\infty} \gamma^t A^{\pi_{\theta}}(\mathbf{s}_t, \mathbf{a}_t) \right]$$

Ignoring distribution mismatch?

$$\sum_t E_{\mathbf{s}_t \sim p_{\theta'}(\mathbf{s}_t)} \left[E_{\mathbf{a}_t \sim \pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t)} \left[\frac{\pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t)}{\pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t)} A^{\pi_{\theta}}(\mathbf{s}_t, \mathbf{a}_t) \right] \right] \stackrel{?}{\approx} \underbrace{\sum_t E_{\mathbf{s}_t \sim p_{\theta}(\mathbf{s}_t)} \left[E_{\mathbf{a}_t \sim \pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t)} \left[\frac{\pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t)}{\pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t)} A^{\pi_{\theta}}(\mathbf{s}_t, \mathbf{a}_t) \right] \right]}_{\bar{A}(\theta')}$$

why do we want this to be true?

$$J(\theta') - J(\theta) \approx \bar{A}(\theta') \quad \Rightarrow \quad \theta' \leftarrow \arg \max_{\theta'} \bar{A}(\theta)$$

2. Use $\hat{A}^{\pi}(\mathbf{s}_t, \mathbf{a}_t)$ to get *improved* policy π'

is it true? and when?

Claim: $p_{\theta}(\mathbf{s}_t)$ is *close* to $p_{\theta'}(\mathbf{s}_t)$ when π_{θ} is *close* to $\pi_{\theta'}$

Bounding the distribution change

Claim: $p_\theta(\mathbf{s}_t)$ is *close* to $p_{\theta'}(\mathbf{s}_t)$ when π_θ is *close* to $\pi_{\theta'}$

Simple case: assume π_θ is a *deterministic* policy $\mathbf{a}_t = \pi_\theta(\mathbf{s}_t)$

$\pi_{\theta'}$ is *close* to π_θ if $\pi_{\theta'}(\mathbf{a}_t \neq \pi_\theta(\mathbf{s}_t) | \mathbf{s}_t) \leq \epsilon$

$$p_{\theta'}(\mathbf{s}_t) = \underbrace{(1 - \epsilon)^t}_{\text{probability we made no mistakes}} p_\theta(\mathbf{s}_t) + (1 - (1 - \epsilon)^t) \underbrace{p_{\text{mistake}}(\mathbf{s}_t)}_{\text{some other distribution}}$$

seem familiar?

$$\frac{1}{2} \sum_{\mathbf{s}_t} |p_{\theta'}(\mathbf{s}_t) - p_\theta(\mathbf{s}_t)| = (1 - (1 - \epsilon)^t) \frac{1}{2} \sum_{\mathbf{s}_t} |p_{\text{mistake}}(\mathbf{s}_t) - p_\theta(\mathbf{s}_t)| \leq 1 - (1 - \epsilon)^t$$

useful identity: $(1 - \epsilon)^t \geq 1 - \epsilon t$ for $\epsilon \in [0, 1]$ $\leq \epsilon t$

not a great bound, but a bound!

Bounding the distribution change

Claim: $p_\theta(\mathbf{s}_t)$ is *close* to $p_{\theta'}(\mathbf{s}_t)$ when π_θ is *close* to $\pi_{\theta'}$

General case: assume π_θ is an arbitrary distribution

$\pi_{\theta'}$ is *close* to π_θ if $\frac{1}{2} \sum_{\mathbf{s}_t} |\pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t) - \pi_\theta(\mathbf{a}_t|\mathbf{s}_t)| = D_{\text{TV}}(\pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t), \pi_\theta(\mathbf{a}_t|\mathbf{s}_t)) \leq \epsilon$

Useful lemma: if $D_{\text{TV}}(p_X(x), p_Y(y)) \leq \epsilon$, exists $p(x, y)$ such that $p(x) = p_X(x)$, $p(y) = p_Y(y)$ and $p(x = y) = 1 - \epsilon$

$\Rightarrow p_X(x)$ “agrees” with $p_Y(y)$ with probability ϵ

$\Rightarrow \pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t)$ takes a different action than $\pi_\theta(\mathbf{a}_t|\mathbf{s}_t)$ with probability at most ϵ

$$\begin{aligned} \frac{1}{2} \sum_{\mathbf{s}_t} |p_{\theta'}(\mathbf{s}_t) - p_\theta(\mathbf{s}_t)| &= (1 - (1 - \epsilon)^t) \frac{1}{2} \sum_{\mathbf{s}_t} |p_{\text{mistake}}(\mathbf{s}_t) - p_\theta(\mathbf{s}_t)| \leq 1 - (1 - \epsilon)^t \\ &\leq \epsilon t \end{aligned}$$

Bounding the objective value

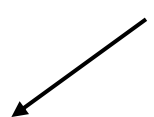
$\pi_{\theta'}$ is *close* to π_{θ} if $\frac{1}{2} \sum_{\mathbf{s}_t} |\pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t) - \pi_{\theta}(\mathbf{a}_t|\mathbf{s}_t)| \leq \epsilon$ for all $\mathbf{s}_t$

$$\frac{1}{2} \sum_{\mathbf{s}_t} |p_{\theta'}(\mathbf{s}_t) - p_{\theta}(\mathbf{s}_t)| \leq \epsilon t$$

$$\begin{aligned} E_{p_{\theta'}(\mathbf{s}_t)}[f(\mathbf{s}_t)] &= \sum_{\mathbf{s}_t} p_{\theta'}(\mathbf{s}_t) f(\mathbf{s}_t) \geq \sum_{\mathbf{s}_t} p_{\theta}(\mathbf{s}_t) f(\mathbf{s}_t) - |p_{\theta}(\mathbf{s}_t) - p_{\theta'}(\mathbf{s}_t)| \max_{\mathbf{s}_t} f(\mathbf{s}_t) \\ &\geq E_{p_{\theta}(\mathbf{s}_t)}[f(\mathbf{s}_t)] - 2\epsilon t \max_{\mathbf{s}_t} f(\mathbf{s}_t) \end{aligned}$$

$$\begin{aligned} \sum_t E_{\mathbf{s}_t \sim p_{\theta'}(\mathbf{s}_t)} \left[E_{\mathbf{a}_t \sim \pi_{\theta}(\mathbf{a}_t|\mathbf{s}_t)} \left[\frac{\pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t)}{\pi_{\theta}(\mathbf{a}_t|\mathbf{s}_t)} A^{\pi_{\theta}}(\mathbf{s}_t, \mathbf{a}_t) \right] \right] &\geq \\ \sum_t E_{\mathbf{s}_t \sim p_{\theta}(\mathbf{s}_t)} \left[E_{\mathbf{a}_t \sim \pi_{\theta}(\mathbf{a}_t|\mathbf{s}_t)} \left[\frac{\pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t)}{\pi_{\theta}(\mathbf{a}_t|\mathbf{s}_t)} A^{\pi_{\theta}}(\mathbf{s}_t, \mathbf{a}_t) \right] \right] &- \sum_t 2\epsilon t C \end{aligned}$$

$O(Hr_{\max})$ or $O\left(\frac{r_{\max}}{1-\gamma}\right)$



maximizing this maximizes a bound on the thing we want!

Where are we at so far?

$$\theta' \leftarrow \arg \max_{\theta'} \sum_t E_{\mathbf{s}_t \sim p_\theta(\mathbf{s}_t)} \left[E_{\mathbf{a}_t \sim \pi_\theta(\mathbf{a}_t|\mathbf{s}_t)} \left[\frac{\pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t)}{\pi_\theta(\mathbf{a}_t|\mathbf{s}_t)} A^{\pi_\theta}(\mathbf{s}_t, \mathbf{a}_t) \right] \right]$$

$$\text{such that } D_{\text{TV}}(\pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t), \pi_\theta(\mathbf{a}_t|\mathbf{s}_t)) \leq \epsilon$$

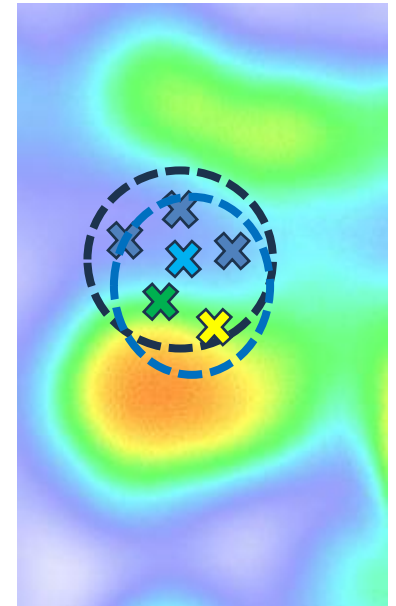
for small enough ϵ , this is guaranteed to improve $J(\theta') - J(\theta)$

PPO with KL divergence

policy gradient with KL-constraint (PPO):

1. sample $\{\tau^{(i)}\}$ from π_θ (run the policy)
2. evaluate $y_t^{(i)} = r(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) + \hat{V}_\phi^\pi(\mathbf{s}_{t+1}^{(i)})$
3. fit $\hat{V}_\phi^\pi(\mathbf{s})$ to targets $\{y_t^{(i)}\}$
4. evaluate $\hat{A}_{\text{GAE}}^\pi(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) = \sum_{t'=t}^{\infty} (\gamma\lambda)^{t'-t} \delta_{t'}^{(i)}$
5. $\theta' \leftarrow \theta$
- $K \times$
 6. $\nabla_{\theta'} J(\theta') \approx \nabla_{\theta'} \mathcal{L}_{\text{KL}}(\theta')$
 7. $\theta' \leftarrow \theta' + \alpha \nabla_{\theta'} J(\theta')$
6. $\beta \leftarrow \beta + \alpha (D_{\text{KL}}(\pi_\theta(\mathbf{a}_t|\mathbf{s}_t) \parallel \pi_{\theta'}(\mathbf{a}_t|\mathbf{s}_t)) - \epsilon)$
7. $\theta \leftarrow \theta'$

$$\mathcal{L}_{\text{KL}}(\theta', \beta) = \sum_{i=1}^N \sum_{t=1}^H \frac{\pi_{\theta'}(\mathbf{a}_t^{(i)}|\mathbf{s}_t^{(i)})}{\pi_\theta(\mathbf{a}_t^{(i)}|\mathbf{s}_t^{(i)})} A^{\pi_\theta}(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) + \beta \log \pi_{\theta'}(\mathbf{a}_t^{(i)}|\mathbf{s}_t^{(i)})$$



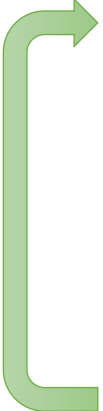
Part 3:

Actor-critic

An online actor-critic algorithm

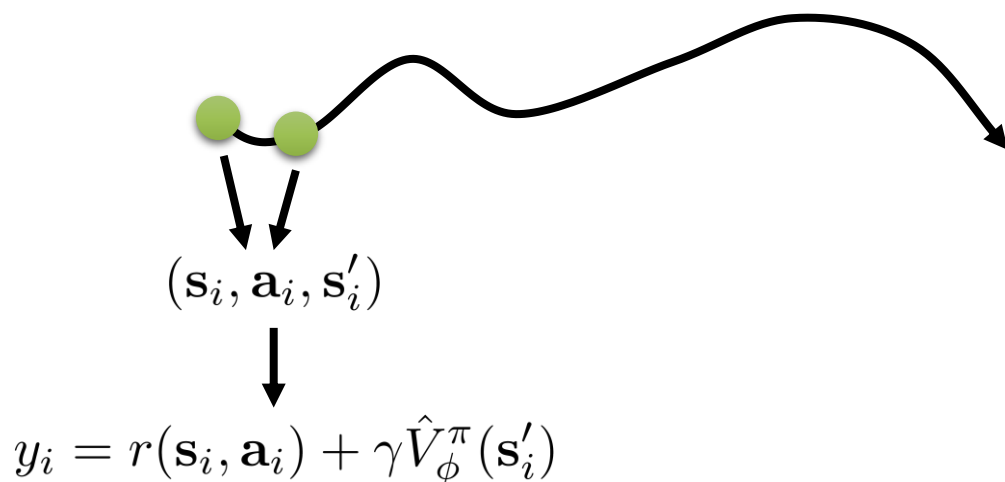
online actor-critic algorithm:

Example algorithm: A3C

- 
1. get $(\mathbf{s}_i, \mathbf{a}_i, \mathbf{s}'_i)$ by taking one step with $\mathbf{a} \sim \pi_\theta(\mathbf{a}|\mathbf{s})$
 2. evaluate $y_i = r(\mathbf{s}_i, \mathbf{a}_i) + \gamma \hat{V}_\phi^\pi(\mathbf{s}'_i)$
 3. update ϕ using $\nabla_\phi \mathcal{L}_i(\phi)$
 4. evaluate $\hat{A}^\pi(\mathbf{s}_i, \mathbf{a}_i) = r(\mathbf{s}_i, \mathbf{a}_i) + \gamma \hat{V}_\phi^\pi(\mathbf{s}'_i) - \hat{V}_\phi^\pi(\mathbf{s}_i)$
 5. $\nabla_\theta J(\theta) \approx \nabla_\theta \log \pi_\theta(\mathbf{a}_i|\mathbf{s}_i) \hat{A}^\pi(\mathbf{s}_i, \mathbf{a}_i)$
 6. $\theta \leftarrow \theta + \alpha \nabla_\theta J(\theta)$

$$\mathcal{L}_i(\phi) = \left\| \hat{V}_\phi^\pi(\mathbf{s}_i) - y_i \right\|^2$$

$$\phi \leftarrow \phi - \alpha \nabla_\phi \mathcal{L}_i(\phi)$$



Eligibility traces & n-step returns

$$\hat{A}_C^\pi(\mathbf{s}_t, \mathbf{a}_t) = r(\mathbf{s}_t, \mathbf{a}_t) + \gamma \hat{V}_\phi^\pi(\mathbf{s}_{t+1}) - \hat{V}_\phi^\pi(\mathbf{s}_t)$$

+ lower variance

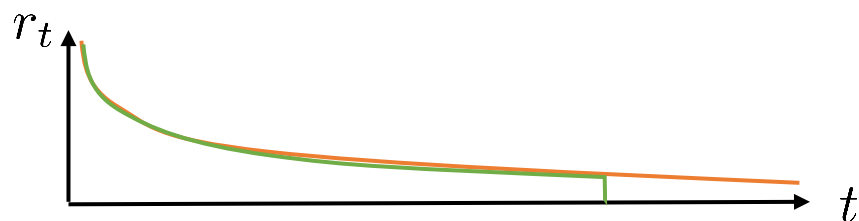
- higher bias if value is wrong (it always is)

$$\hat{A}_{MC}^\pi(\mathbf{s}_t, \mathbf{a}_t) = \sum_{t'=t}^{\infty} \gamma^{t'-t} r(\mathbf{s}_{t'}, \mathbf{a}_{t'}) - \hat{V}_\phi^\pi(\mathbf{s}_t)$$

+ no bias

- higher variance (because single-sample estimate)

Can we combine these two, to control bias/variance tradeoff?



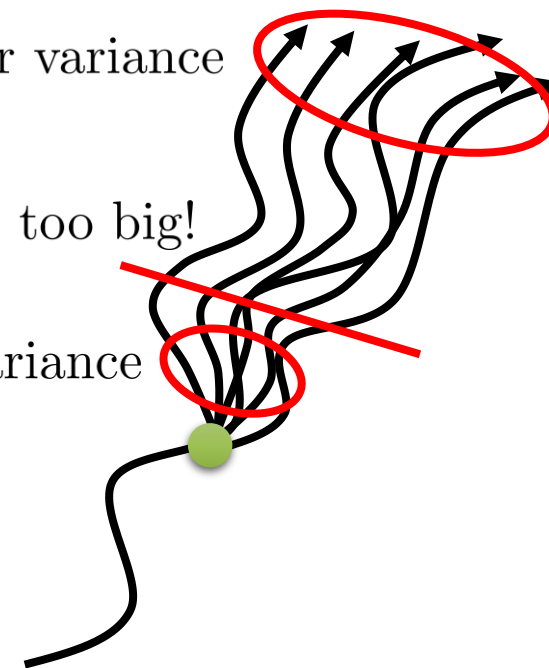
$$\hat{A}_n^\pi(\mathbf{s}_t, \mathbf{a}_t) = \sum_{t'=t}^{t+n} \gamma^{t'-t} r(\mathbf{s}_{t'}, \mathbf{a}_{t'}) - \hat{V}_\phi^\pi(\mathbf{s}_t) + \gamma^n \hat{V}_\phi^\pi(\mathbf{s}_{t+n})$$

choosing $n > 1$ often works better!

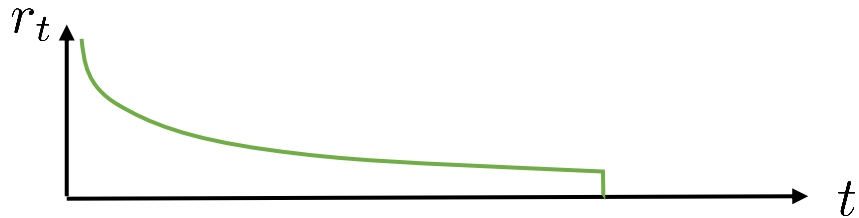
bigger variance

cut here before variance gets too big!

smaller variance



Generalized advantage estimation



Do we have to choose just one n ?

Cut everywhere all at once!

$$\hat{A}_n^\pi(\mathbf{s}_t, \mathbf{a}_t) = \sum_{t'=t}^{t+n} \gamma^{t'-t} r(\mathbf{s}_{t'}, \mathbf{a}_{t'}) - \hat{V}_\phi^\pi(\mathbf{s}_t) + \gamma^n \hat{V}_\phi^\pi(\mathbf{s}_{t+n})$$

$$\hat{A}_{\text{GAE}}^\pi(\mathbf{s}_t, \mathbf{a}_t) = \sum_{n=1}^{\infty} w_n \hat{A}_n^\pi(\mathbf{s}_t, \mathbf{a}_t)$$

weighted combination of n -step returns

How to weight?

Mostly prefer cutting earlier (less variance)

$w_n \propto \lambda^{n-1}$ exponential falloff

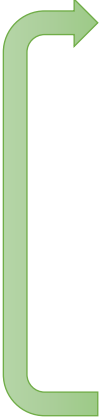
$$\hat{A}_{\text{GAE}}^\pi(\mathbf{s}_t, \mathbf{a}_t) = r(\mathbf{s}_t, \mathbf{a}_t) + \gamma((1 - \lambda)\hat{V}_\phi^\pi(\mathbf{s}_{t+1}) + \lambda(r(\mathbf{s}_{t+1}, \mathbf{a}_{t+1}) + \gamma((1 - \lambda)\hat{V}_\phi^\pi(\mathbf{s}_{t+2}) + \lambda r(\mathbf{s}_{t+2}, \mathbf{a}_{t+2}) + \dots))$$

$$\hat{A}_{\text{GAE}}^\pi(\mathbf{s}_t, \mathbf{a}_t) = \sum_{t'=t}^{\infty} (\gamma\lambda)^{t'-t} \delta_{t'} \quad \delta_{t'} = r(\mathbf{s}_{t'}, \mathbf{a}_{t'}) + \gamma\hat{V}_\phi^\pi(\mathbf{s}_{t'+1}) - \hat{V}_\phi^\pi(\mathbf{s}_{t'})$$

← similar effect as discount!

Policy gradient with GAE

policy gradient with GAE:

- 
1. sample $\{\tau^{(i)}\}$ from π_θ (run the policy)
 2. evaluate $y_t^{(i)} = r(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) + \gamma \hat{V}_\phi^\pi(\mathbf{s}_{t+1}^{(i)})$
 3. fit $\hat{V}_\phi^\pi(\mathbf{s})$ to targets $\{y_t^{(i)}\}$
 4. evaluate $\hat{A}_{\text{GAE}}^\pi(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) = \sum_{t'=t}^{\infty} (\gamma\lambda)^{t'-t} \delta_{t'}^{(i)}$
 5. $\nabla_\theta J(\theta) \approx \sum_i \sum_t \nabla_\theta \log \pi_\theta(\mathbf{a}_t^{(i)} | \mathbf{s}_t^{(i)}) \hat{A}^\pi(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)})$
 6. $\theta \leftarrow \theta + \alpha \nabla_\theta J(\theta)$

can use GAE estimator here too!

Other good options:

use n -step return estimators

use Monte Carlo estimator with $\hat{V}^\pi(\mathbf{s}_t)$ baseline

in practice, we typically
center the advantages

$$\mu = \frac{1}{HN} \sum_{i,t} \hat{A}^\pi(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)})$$

$$\sigma = \sqrt{\frac{1}{HN} \sum_{i,t} \left(\hat{A}^\pi(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) - \mu \right)^2}$$

$$\bar{A}^\pi(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) = \frac{\hat{A}^\pi(\mathbf{s}_t^{(i)}, \mathbf{a}_t^{(i)}) - \mu}{\sigma}$$

This is the first part of implementing PPO

The other part is a slightly different gradient estimator

Off-policy actor-critic with reparameterization

off-policy reparameterized actor-critic algorithm:

-
1. get $(\mathbf{s}_i, \mathbf{a}_i, \mathbf{s}'_i)$ by taking one step with $\mathbf{a} \sim \pi_\theta(\mathbf{a}|\mathbf{s})$
 2. evaluate $y_i = r(\mathbf{s}_i, \mathbf{a}_i) + \gamma \mathbb{E}_{\mathbf{a}' \sim \pi_\theta(\mathbf{a}'|\mathbf{s}'_i)} [\hat{Q}_\phi^\pi(\mathbf{s}'_i, \mathbf{a}')]]$
 3. update ϕ using $\nabla_\phi \sum_{i=1}^B \|\hat{Q}_\phi^\pi(\mathbf{s}_i, \mathbf{a}_i) - y_i\|^2$
 4. $\nabla_\theta J(\theta) \approx \sum_i \nabla_\theta \hat{Q}_\phi^\pi(\mathbf{s}, \mu_\theta(\mathbf{s}) + \sigma_\theta(\mathbf{s})\epsilon_i)$
 5. $\theta \leftarrow \theta + \alpha \nabla_\theta J(\theta)$

$$\begin{aligned}\mathbb{E}_{\mathbf{a} \sim \pi_\theta(\mathbf{a}|\mathbf{s})} [Q(\mathbf{s}, \mathbf{a})] &= \mathbb{E}_{\epsilon \sim \mathcal{N}(0,1)} [Q(\mathbf{s}, \mu_\theta(\mathbf{s}) + \sigma_\theta(\mathbf{s})\epsilon)] \\ &\approx Q(\mathbf{s}, \mu_\theta(\mathbf{s}) + \sigma_\theta(\mathbf{s})\epsilon) \quad \epsilon \sim \mathcal{N}(0,1)\end{aligned}$$